USING STATISTICAL MODELS AND MACHINE LEARNING TO FORECAST PRODUCT DEMAND

Student Project for MIS 495: Strategic Business Analytics

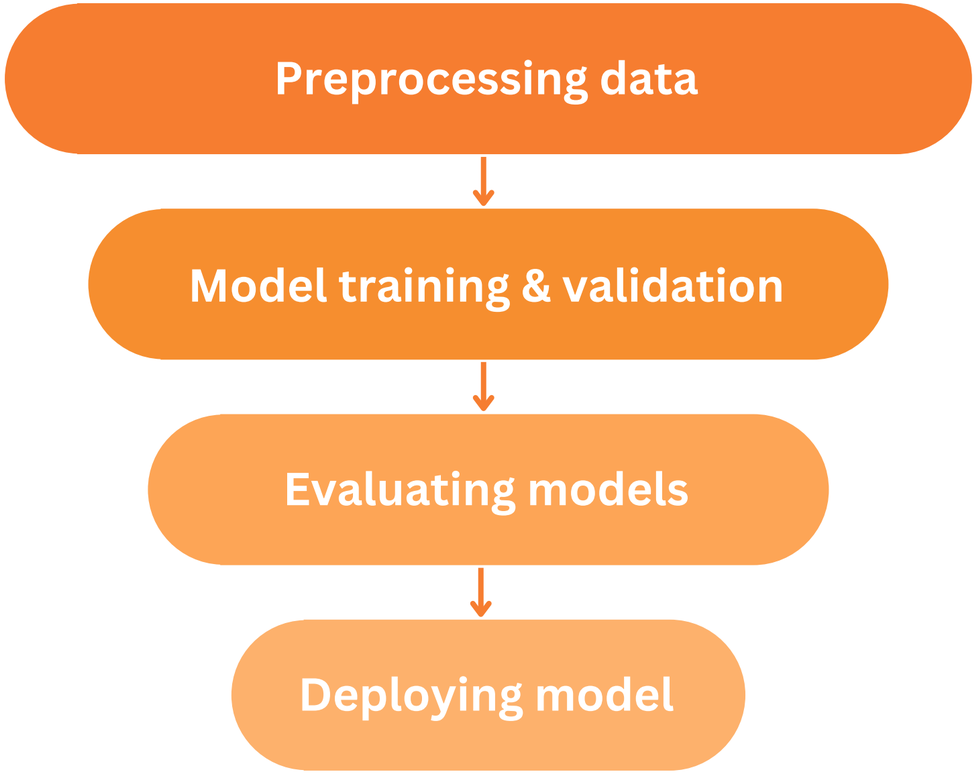
Presentation | 20th March 2025

Background context and flow of work

- This report focuses on demand forecasting for Ashley's outdoor and accessories product line, with the primary objective of **assessing the accuracy of potential forecasting models.**

- The work aims to **optimize inventory planning strategies**, ensuring that supply chain management operates efficiently, supporting better decision-making, and achieving cost efficiency across the business.

Approach and flow

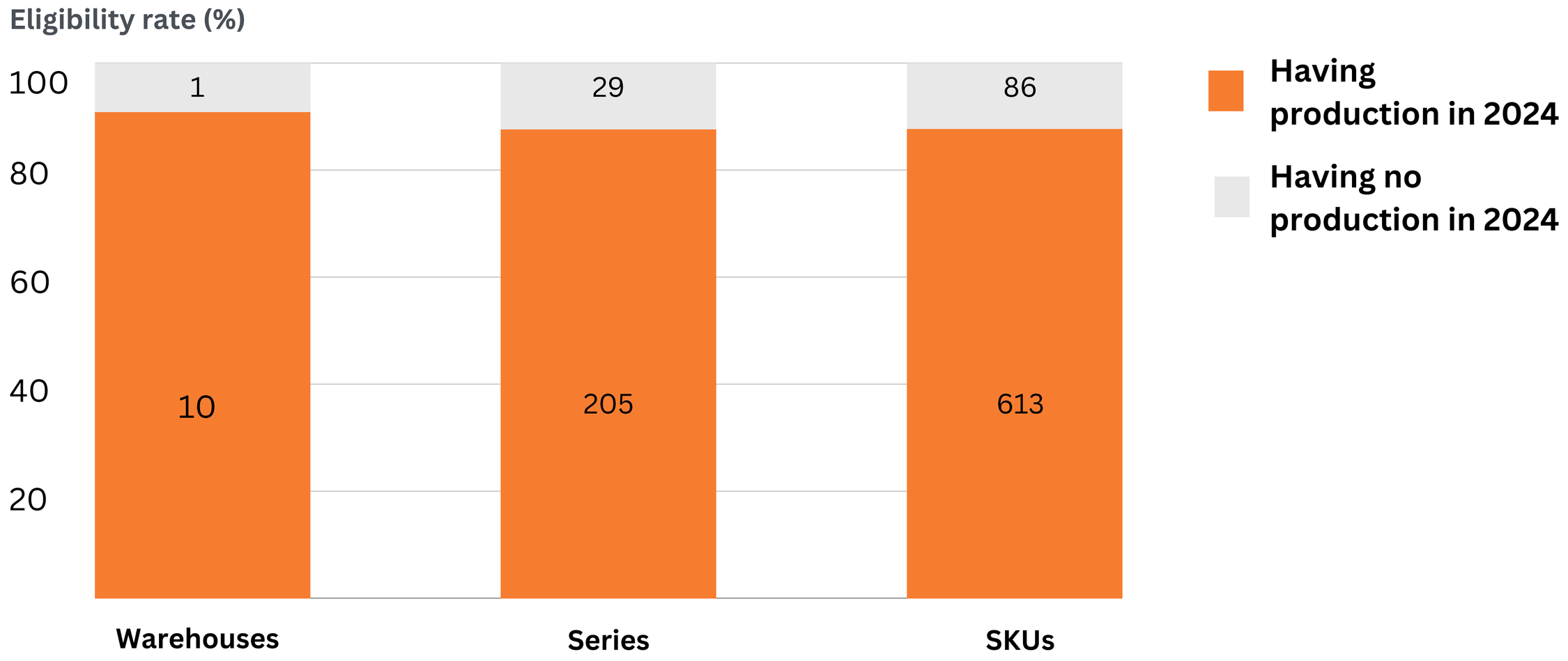


Executive summary

- **Current situation:**
  - Ashley Furniture relies on traditional forecasting method (e.g., moving average).
  - A efficient and accurate machine learning forecasting model is crucial.
  - Optimize inventory levels and enhance operational efficiency
- **Results:**
  - Forecasting demand for outdoor (seasonal) and accessory products for Q1 2025 using 3 years of historical data.
  - Successful models created for 36% of Outdoor series and 14% of Accessories series across all warehouses.
- **Challenge:**
  - A warehouse, 28 series, and 86 SKUs of outdoor products lack data for 2024, complicating the 2025 forecast for these products.
- **Solution:**
  - Store & Wait – Hold the 2024 stop-production data until sales data becomes available again.

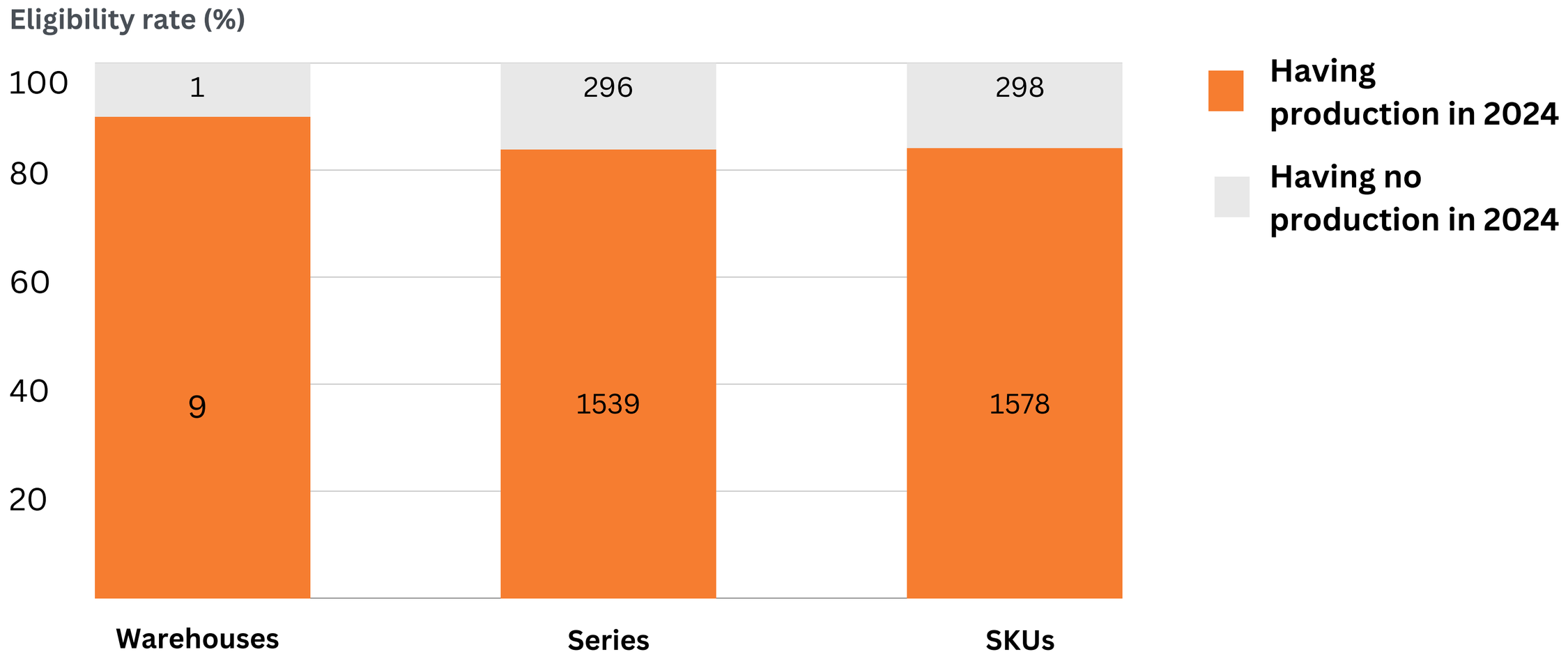
Build forecast models on 10 warehouses, 205 series, and 613 SKUs of outdoor product

Only records with production in 2024 are used



Build forecast models on 9 warehouses, 1539 series, and 613 SKUs of accessories product

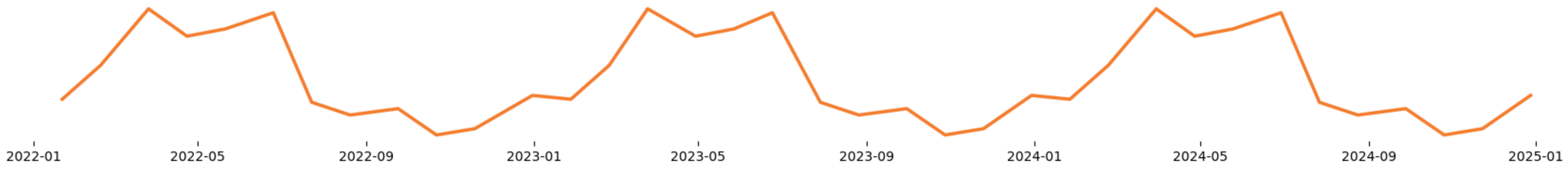
Only records with production in 2024 are used



Seasonality in outdoor products versus stable demand for accessories

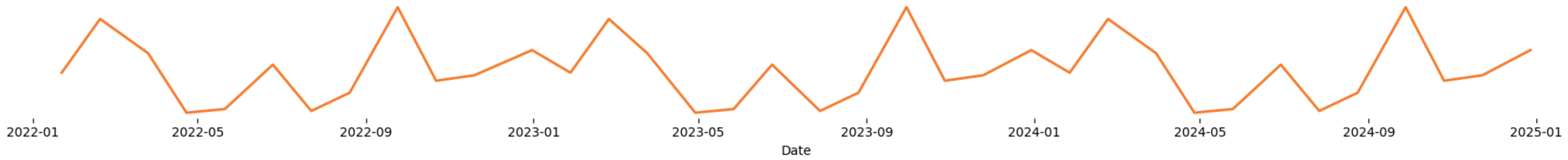
Demand for outdoor products peaks in summer, drops in winter

Seasonality of outdoor product demand



No clear trend or seasonality in accessories products

Seasonality of accessory product demand



6 models are deployed for 3 series types across 2 product lines

X represents stable demand, Y represents mildly fluctuating demand, and Z represents highly volatile demand

	Outdoor products	Accessory products
Series Type	**Models**	**Models**
X	Holt-Winters & ARIMA	ARIMA, Holt-Winters
Y	Holt-Winters & ARIMA	XGBoost, ARIMA, Holt-Winters
Z	XGBoost & Neural Network Ensemble	XGBoost, ARIMA, Holt-Winters

Models achieve the target on 73 outdoor series and 216 accessories series

Assessing error thresholds and target achievement for outdoor and accessories products

		Outdoor products	Accessories products
Series type	**Error threshold (%)**	**% of models meeting target**	**% of models meeting target**
X	**Less than 25%**	01 / 02 (50%)	75 / 124 (60%)
Y	**Less than 30%**	27 / 73 (37%)	122 / 442 (28%)
Z	**Less than 35%**	44 / 131 (34%)	19 / 974 (1,6%)

Tailored Modeling is Key to Capturing Demand Patterns at Levels Y and Z

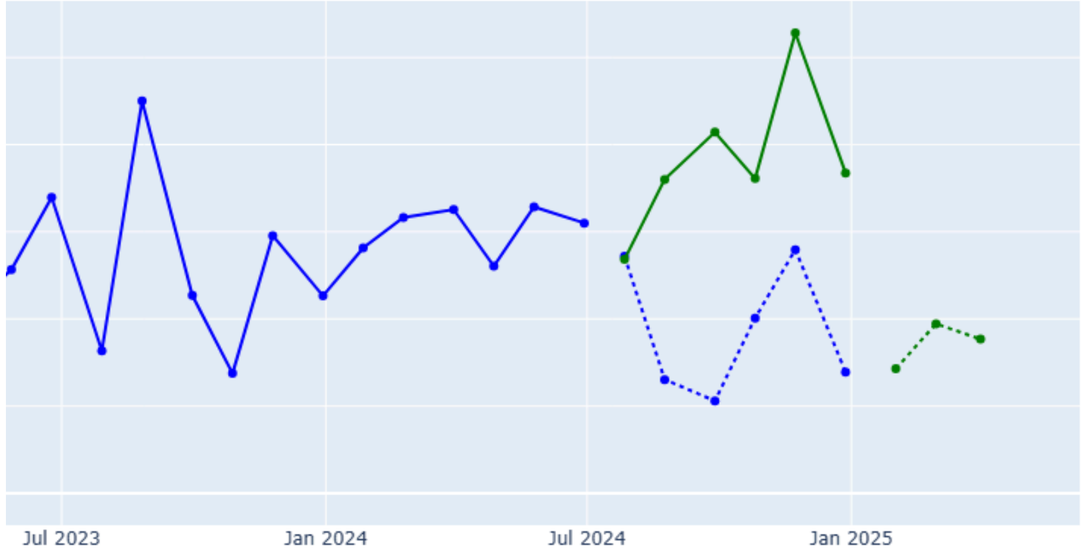
High Demand Fluctuations in Levels Y and Z Reduce Model Accuracy

- In levels Y and Z, fluctuations occur due to the emergence of many new product lines in 2023 and 2024, making model training difficult due to limited data

- When shifting to levels Y and Z, the difference in volatility makes it impossible for a single machine learning model to accurately forecast demand for all series in these levels.

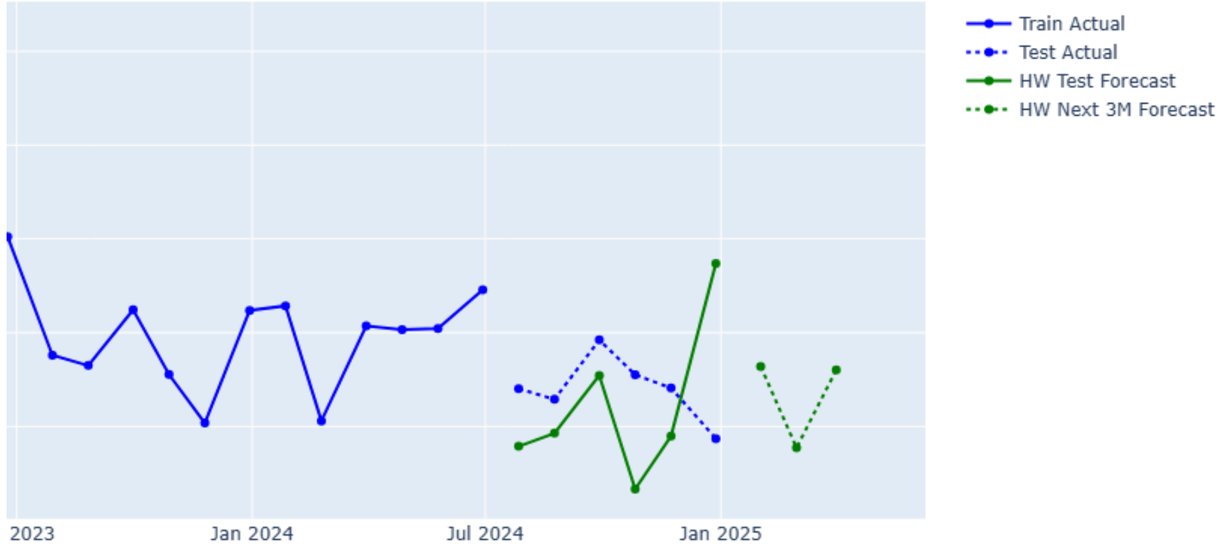
Varying performance of the Holt-Winters model on stable demand series in outdoor products

A single model struggles to perform well across all series



Series: P309

wMAPE: 98%

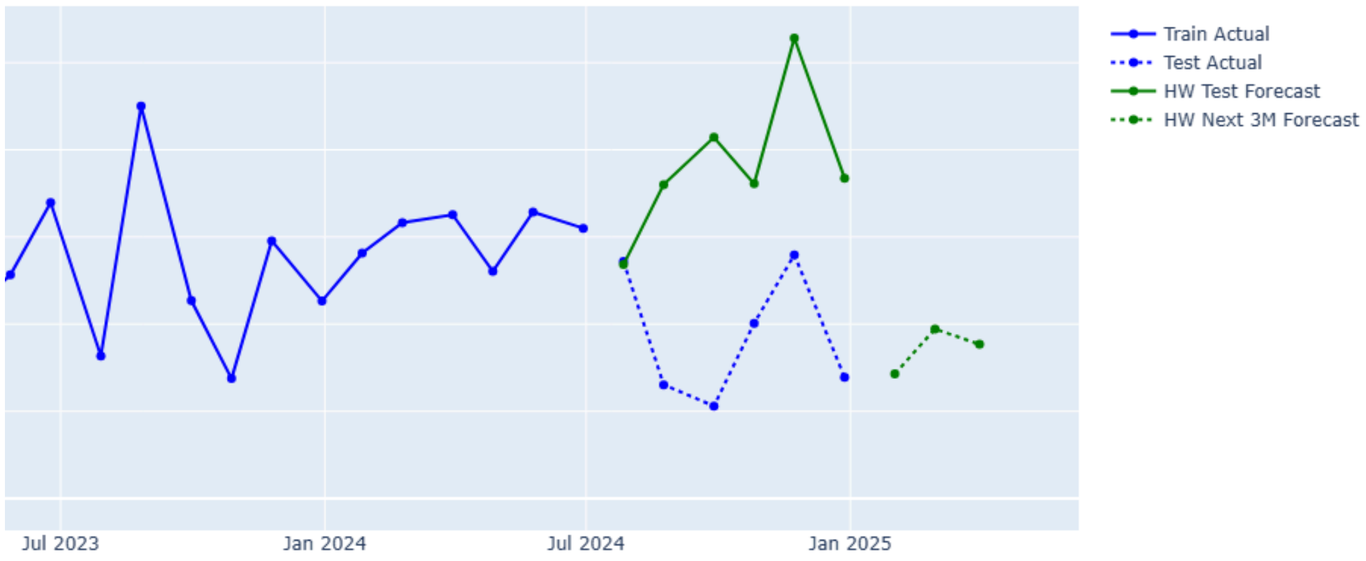


Series: P801

wMAPE: 25%

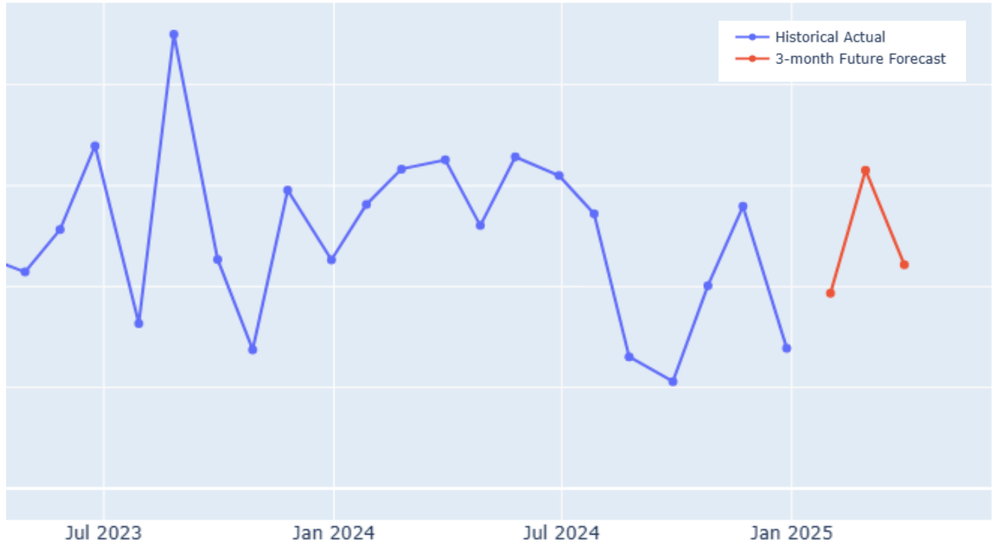
Enhancing forecast accuracy with deep learning for previously underperforming models

Proving the feasibility of an improved model by using LSTM



Series: P309 (before)

wMAPE: 98%



Series: P309 (after)

wMAPE: 17.25%

Conclusion

By adding **Neural Networks** to **machine learning models**, even for series with high errors, we can still continue to optimize and **reduce WMAPE**. However, this optimization is **limited by the current computational resources**. A feasible solution is to use ensemble models combining both neural network and machine learning for different product lines to improve forecast accuracy.

Future work

Independent Execution: To significantly enhance forecast accuracy and minimize errors, it is strongly recommended that Ashley's team leverage the abundant computational resources to implement a robust machine learning and deep learning approach, actively integrating these advanced techniques to optimize WMAPE and drive substantial improvements in forecasting performance

Scan to view demand forecast for quarter 1, 2025

Forecast demand for the first quarter in 2025– Gradio web interface

Outdoor Products

Accessories Products

THANKS FOR LISTENING